

The weight $\omega_\ell^g = \sum_e \omega_{e,\ell}^g$ sums over the weights $\omega_{e,\ell}^g$ from Proposition 1, and is equal to the population regression coefficient from the following auxiliary regression:

$$D_{i,t}^\ell = \alpha_i + \lambda_t + \sum_{g \in \mathcal{G}} \omega_{e,\ell}^g \mathbf{1}\{t - E_i \in g\} + v_{i,t} \quad (17)$$

which regresses $D_{i,t}^\ell$ on all bin indicators included in regression (4) and two-way fixed effects.

We note that even under treatment effect homogeneity μ_ℓ can still be contaminated by treatment effects from the excluded periods. This contamination, however, can be avoided by adjusting the specification to only exclude periods with zero treatment effect.

For specifications with relative time bins, we note that there can still be contamination from other bins as suggested by the second term of expression (16). A sufficient condition to avoid such contamination would be to group relative periods ℓ' into a bin only when their effects are the same since their weights $\omega_{\ell'}^g$ would sum to zero.

For the “fully dynamic” specification (7) where all g ’s are singletons of relative time periods, the weight $\omega_{\ell'}^\ell$ is zero for each relative period $\ell' \neq \ell$ that is included in the specification. The decomposition therefore simplifies to

$$\mu_\ell = ATT_\ell + \sum_{\ell' \in g^{excl}} \omega_{\ell'}^g ATT_{\ell'}. \quad (18)$$

3.6 Intuition for contamination

Proposition 3 demonstrates that even under the assumptions of parallel trends and no anticipation, estimates μ_g can still be contaminated by treatment effects from other periods. In this section we explain the intuition behind why this contamination occurs for the “fully dynamic” specification (7). A decomposition of μ_ℓ into a weighted average of $CATT_{e,\ell'}$ demonstrates that contamination is driven by the interaction of two elements: the weights and $CATT_{e,\ell'}$. The weights underlying the contamination are non-linear functions of the distribution of the cohorts. We do not attempt to provide a heuristic for determining the magnitude of the weights, but instead describe how to estimate the weights and later on in Section 4 how to estimate each $CATT_{e,\ell}$. This allows researchers to directly determine the degree of contamination in their application. Our publicly-available Stata package `eventstudyweights` automates the estimation of these weights using the panel dataset underlying any given specification of (1).

We apply the familiar omitted variable bias (OVB) formula to arrive at our decomposition. In an event study where individuals receive the treatment at different times, the panel can never be balanced in both calendar time and time relative to the initial treatment. As a result, the relative time indicators are still correlated even after controlling for unit and time fixed effects in a two-way fixed effects regression. We use the saturated regression and the OVB formula to illustrate how this correlation leads to contamination. We defer its formal derivation to Appendix B.

Under the parallel trends assumption only, the saturated regression is

$$\begin{aligned}
Y_{i,t} = & \sum_e E[Y_{i,0}^\infty | E_i = e] \cdot \mathbf{1}\{E_i = e\} + \sum_s E[Y_{i,s}^\infty - Y_{i,0}^\infty] \cdot \mathbf{1}\{t = s\} \\
& + \sum_{\ell' \in g^{incl}} \sum_{e \in I_{\ell'}} CATT_{e,\ell'} \cdot \left(D_{i,t}^{\ell'} \cdot \mathbf{1}\{E_i = e\} \right) \\
& + \sum_{\ell' \in g^{excl}} \sum_{e \in I_{\ell'}} CATT_{e,\ell'} \cdot \left(D_{i,t}^{\ell'} \cdot \mathbf{1}\{E_i = e\} \right) + \epsilon_{i,t}
\end{aligned} \tag{19}$$

where the regressors are cohort fixed effects, time fixed effects, and cohort-specific relative time indicators. Furthermore, let g^{incl} collect the relative time included in (7). The coefficient associated with the cohort-specific relative time indicator $D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}$ is the cohort-specific average treatment effects $CATT_{e,\ell}$. To decompose the coefficient μ_ℓ from (7) in terms of this saturated regression (19), the OVB formula multiplies the coefficients in the saturated regression (19), $CATT_{e,\ell'}$, with the regression coefficients from (12), $\omega_{e,\ell'}^\ell$, which leads to the following decomposition for μ_ℓ as a linear combination of $CATT_{e,\ell'}$:

$$\mu_\ell = \sum_{e,\ell'} \omega_{e,\ell'}^\ell CATT_{e,\ell'}. \tag{20}$$

Since $\omega_{e,\ell'}^\ell$ is equal to a regression coefficient from (12), we can write it as

$$\omega_{e,\ell'}^\ell = (\sigma_{e,\cdot})^\top \mathbf{\Delta}_{e+\ell'} A_\ell^{-1}. \tag{21}$$

Below we briefly comment on each of the three elements in the above expression to highlight how they depend on the distribution of the cohorts. We defer their detailed definitions and derivations to Appendix A.1.

- $\sigma_{e,\cdot}$ is a vector of the covariance between cohort e and the other cohorts, namely $Cov(\mathbf{1}\{E_i = e\}, \mathbf{1}\{E_i = e'\})$.

This term thus scales quadratically in the share of cohort e , and is small for small cohorts.

- Δ_t is a matrix of demeaned relative time indicators. The entry that corresponds to cohort e' and relative time indicator $D_{i,t}^\ell$ is $E[D_{i,t}^\ell \mid E_i = e'] - \frac{1}{T+1} \mathbf{1}\{e' \in \mathcal{I}_\ell\}$. When T is large, i.e. the panel is long, the second term is small and this entry is therefore approximately equal to the relative time indicator.
- A_ℓ^{-1} is the row of A^{-1} that corresponds to the relative time indicator $D_{i,t}^\ell$ for A the covariance matrix of demeaned relative time indicators. Specifically, the entry in A that corresponds to the covariance between demeaned $D_{i,t}^\ell$ and $D_{i,t}^{\ell'}$ is

$$\sum_t Cov(D_{i,t}^\ell, D_{i,t}^{\ell'}) - \frac{1}{T+1} Cov(\mathbf{1}\{E_i \in \mathcal{I}_\ell\}, \mathbf{1}\{E_i \in \mathcal{I}_{\ell'}\}). \quad (22)$$

Within any time period $D_{i,t}^\ell$ and $D_{i,t}^{\ell'}$ are negatively correlated because no cohorts can be in these two relative times at the same time. The second covariance term is in general also non-zero because being in one cohort predicts (not) being in another cohort. Therefore A is in general not a diagonal matrix and A^{-1} would depend on the distribution of the cohorts non-linearly.

The three elements of (21) demonstrate the weights are non-linear functions of the distribution of the cohorts, and they are in general non-zero. Nonetheless these weights can be estimated easily by the auxiliary regression (12).

3.7 Invalidity of pretrend tests based on pre-period coefficients.

Contamination undermines the practice of testing for pretrends using pre-period coefficients. Proposition 3 implies that when effects are not homogenous across cohorts, it is problematic to interpret non-zero estimates for μ_g as evidence for pretrends, where the set g contains some leads $\ell < 0$. Proposition 4 implies that even with homogeneous treatment effect, if the effects associated with the excluded periods are not zero, then contamination may still occur. Therefore without strong assumptions, pre-period coefficients should not be used to test for pretrends because contamination can lead to estimates that are non-zero in the absence of pretrends or zero in the presence of pre-trends.

Testing for pretrends using pre-period coefficients is commonly used in practice. As an example, He and Wang (2017) mention “the estimated coefficients of the leads of treatments, i.e. δ_k for all $k \leq -2$

are statistically indifferent from zero” as evidence for lack of pretrends. As another example, Chetty et al. (2014) assert “there is no trend toward higher individual pension contributions prior to year 0 ... as one would expect if individuals’ tastes for saving were changing around the job switch” based on pre-period coefficient estimates. These tests are only appropriate when the authors are willing to make strong assumptions.

To provide further intuition for why this test is not meaningful without additional assumptions we walk through a simple example of the fully dynamic specification. Consider a balanced panel with $T = 2$ and cohorts $E_i \in \{1, 2\}$. There are at least two multi-colinearities from including all four relative time indicators. To form the fully dynamic specification we include $g^{incl} = \{-2, 0\}$ and exclude $g^{excl} = \{-1, 1\}$:

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_{\ell \in \{-2, 0\}} \mu_\ell D_{i,t}^\ell + \nu_{i,t}. \quad (23)$$

The choice of g^{excl} is based on the common practice of normalizing relative to the -1 period and distant lags.

When there are no never treated units, we can express the pre-trend coefficient μ_{-2} in terms of *CATTs*:

$$\mu_{-2} = \underbrace{CATT_{2,-2}}_{\text{own period}} + \underbrace{\frac{1}{2}CATT_{1,0} - \frac{1}{2}CATT_{2,0}}_{\ell' \in g^{incl}, \ell' \neq -2} + \underbrace{\frac{1}{2}CATT_{1,1} - CATT_{1,-1} - \frac{1}{2}CATT_{2,-1}}_{\ell' \in g^{excl}} \quad (24)$$

It is apparent the weights maintain the structure described in Proposition 1. Without any anticipation effect, the effects $CATT_{e,\ell < 0}$ are zero and thus we expect μ_{-2} to be zero regardless of the cohort shares. With homogeneous treatment effect, cohorts 1 and 2 experience the same treatment effect at relative time 0 so that $CATT_{1,0}$ and $CATT_{2,0}$ cancel. But even with homogeneous treatment effect, the last term reflects the role of excluded periods as $CATT_{1,-1}$, $CATT_{2,-1}$ and $CATT_{1,1}$ receive non-zero weights. If there is any lagged effect and $CATT_{1,1}$ is non-zero, the coefficient μ_{-2} would be non-zero even without any anticipation effect. Note such behavior is independent of the distribution of the two cohorts.

We can further introduce never treated units to our example to demonstrate how the weight $\omega_{e,\ell'}^{-2}$ can be non-linear in the distribution of cohorts while maintaining the structure described in Proposition 1. In Figure 1 we plot the weights $\omega_{e,\ell'}^{-2}$ as we vary the distribution of cohorts. Specifically, we vary the total share of treated cohorts (shown on the x -axis), holding the shares of cohort 1 and 2 equal to each other and setting the remaining to be the share of never treated units. For any distribution of cohorts, we have $\omega_{2,-2}^{-2} = 1$ (not pictured). Panel (a) shows the weights for the included period $\ell' = 0$ while panel (b) shows the weights for